Data Mining & Machine Learning

M.Sc. in Artificial Intelligence and Data Engineering

University of Pisa

Project Report

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Project Title: Classification of Human Body Postures thorugh Wearable Sensors

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Abstract

This study investigates the automated recognition of human body postures using wearable accelerometer data. A labelled multi-subject dataset was employed to formulate the problem as a multi-class classification task. The proposed methodology includes data preprocessing, feature engineering, model training, and evaluation using several machine learning algorithms, namely Decision Trees, k-Nearest Neighbors (KNN), Random Forest, Naïve Bayes, and AdaBoost. Model performance was assessed through a nested cross-validation framework with Leave-One-Subject-Out (LOSO) testing to evaluate generalization across unseen subjects.

The results indicate that appropriate data preprocessing and feature engineering significantly improve classification accuracy. However, the LOSO evaluation highlights limited generalization capability when models are applied to previously unseen users. Overall, the findings demonstrate that wearable accelerometers can effectively support posture recognition, despite the use of a small dataset comprising only four subjects and a low sampling frequency (8 Hz). The main challenge remains the accurate classification of transitional movements between postures.

1. Introduction

Human Activity Recognition (HAR) has become an important research area due to its potential applications in healthcare, ambient assisted living, rehabilitation, and elderly care. As life expectancy increases and populations age, the development of technologies capable of monitoring human behavior and supporting independent living has gained significant social and economic relevance. Recent studies highlight the growing importance of HAR systems in supporting frail individuals, enabling continuous monitoring, and improving the quality of healthcare services while reducing associated costs [1]. In particular, recognizing body postures and daily activities can provide valuable information for patient monitoring, fall prevention, and personalized healthcare interventions.

Among the various approaches to HAR, wearable sensors have emerged as a practical and cost-effective solution. Compared with vision-based systems, wearable sensors are less intrusive, preserve users' privacy, and can operate in a wide range of environments. Accelerometers are especially popular due to their low cost, low power consumption, and ability to capture body movements with sufficient accuracy for activity recognition tasks. Recent reviews confirm that wearable sensor-based systems represent one of the most widely adopted technologies for monitoring human motion and posture in both clinical and real-world settings [2].

The problem addressed in this project is the automatic classification of human body postures using data collected from wearable accelerometers. The dataset consists of measurements acquired from multiple subjects performing both static postures and transition movements. This problem can be formulated as a multi-class classification task, where the objective is to correctly identify the posture associated with each observation. Although previous studies have reported promising results, the ability of machine learning models to generalize to unseen users remains a challenging issue, particularly when datasets contain a limited number of participants. A recent systematic review identified cross-subject generalization and robustness to inter-person variability as some of the main open challenges in HAR research [3].

To investigate this problem, several machine learning algorithms are evaluated within a rigorous experimental framework. Particular attention is given to data preprocessing, feature engineering, model selection, and validation through Leave-One-Subject-Out (LOSO) cross-validation, which provides a realistic assessment of performance on unseen individuals. This evaluation strategy is particularly relevant because it directly addresses one of the key challenges highlighted in the recent HAR literature, namely the development of models capable of maintaining high performance when deployed on users not represented in the training data.

This project aims to:

1. develop a predictive model for human posture and movements recognition;
2. compare the performance of multiple model families: Decision Tree, Random Forest, AdaBoost, KNN, Naïve Bayes;
3. evaluate the generalization capability of the proposed models using a nested Leave-One-Subject-Out cross-validation framework and analyze the impact of feature engineering on classification performance;
4. provide global and local model explanations using SHAP.

2. Problem

Table 1 summarizes the problem and dataset.

Table 1. Problem and dataset at a glance.

Attribute	Summary
Task type	Multi-class classification
Dataset	UCI Wearable Computing: Classification of Body Postures and Movements (PUC-Rio) — 165,632 samples, 18 features
Target variable	class (sitting-down, standing-up, standing, walking and sitting)
Features	18 features: 16 numerical, 2 categorical — see Section 2.2
Missing values	No missing values, one malformed cell
Class distribution	sitting 30.57% · standing 28.60% · walking 26.20% · standingup 7.50% · sittingdown 7.14%
Evaluation metric(s)	F1-score Train / Val / Test Held-out user: 3 users / 3-fold / 1 user — stratified

2.1 Exploratory Data Analysis

Analysis has been conducted on a dataset of 165,000 samples of 4 users (described in Table 2). Dataset class balancing is shown in Figure 1. Static and cyclic classes dominate against transitional classes. In the dataset there are no missing values, just one malformed row.

user	age	height	weight	BMI
debora	46.0	1.62	75.0	28.6
jose_carlos	75.0	1.67	67.0	24.0
katia	28.0	1.58	55.0	22.0
wallace	31.0	1.71	83.0	28.4

Table 2. Users anthropometric data

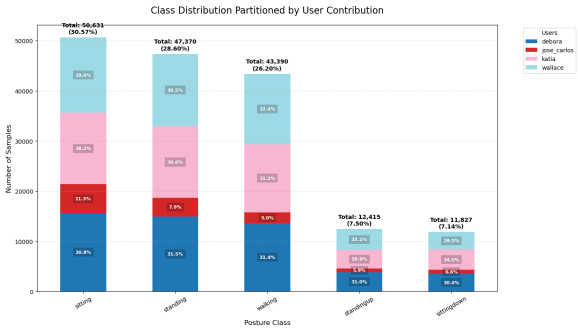


Figure 1. Class balance across posture and movement labels.

2.1.1 Raw Features

Figures 2 and 3 show sensor 2 (on the left thigh) samples, representative of the general behaviour of all the accelerometers. It can be observed that dynamic activities like walking have a significantly higher variance compared to static positions (sitting and standing). Whatsmore the sensor displays amplitude peaks while walking whereas transition activities have skewed shape, as their nature suggests. Correlation is stronger within the same sensor, and around zero between different sensors. Figure 4 is a good approximation of what happens in heatmaps grouped by class.

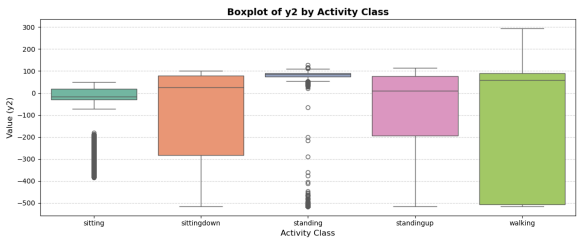


Figure 2. Boxplot of raw sensor values grouped by class.

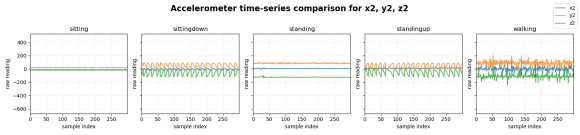


Figure 3. Time-series view of sensor measurements.

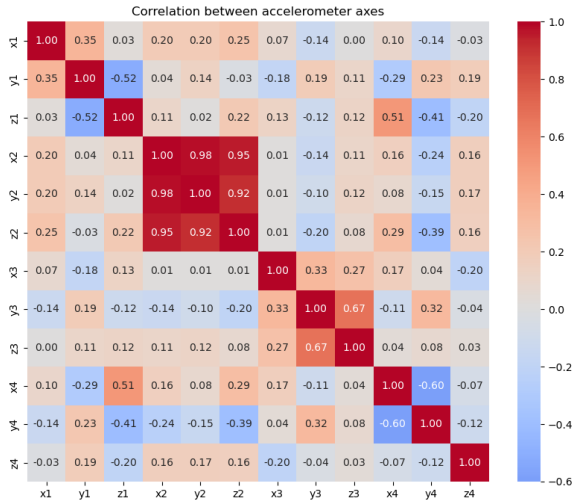


Figure 4. Correlation heatmap of numerical features.

2.1.2 Engineered Features

The dataset has been augmented in two different directions.

- Magnitude, Roll, Yaw and Pitch of each sensor.
- Statistical description of windows.

2.2 Feature Description

The original dataset provides subject attributes and raw tri-axial accelerometer readings collected from four body positions: waist (sensor 1), left thigh (sensor 2), right ankle (sensor 3), and right upper arm (sensor 4) [4]. The raw signals are denoted as x_i, y_i, and z_i for sensor $i \in \{1, 2, 3, 4\}$ and are sampled at $8\,\mathrm{Hz}$. The most useful engineered representation was

obtained by grouping consecutive samples into windows and computing simple statistics for each signal. This produced 48 window-based features and improved classification performance, whereas magnitude and orientation features did not provide a clear advantage over the original Cartesian accelerometer coordinates. Features are described in Table 3.

Group	Features	Role
Subject attributes	User, gender, age, height, weight, BMI	Describe inter-subject variability and body characteristics that can affect acceleration patterns.
Raw sensors	x_i, y_i, z_i for $i = 1, \ldots, 4$	Capture body motion along the three local axes for each wearable sensor.
Orientation and intensity	Magnitude$_i$, roll$_i$, pitch$_i$, yaw$_i$	Summarize movement intensity and spatial orientation; experiments showed limited additional benefit.
Window statistics	Mean$_j$, variance$_j$, min$_j$, max$_j$	Describe short temporal windows and improve discrimination between static and dynamic activities.

Table 3. Summary of original and engineered feature groups.

3. Pipeline and Experiments

The methodology separates dataset optimization from classifier optimization. First, a Decision Tree proxy model was used to compare dataset variants and select the most effective feature representation. Then, the selected Windowed dataset was used to train and compare Decision Tree, Random Forest, AdaBoost, k-NN, and Naive Bayes under the same nested LOSO validation scheme. Figure 5 summarizes the search process, while Figure 6 shows the evaluation pipeline used for each model.

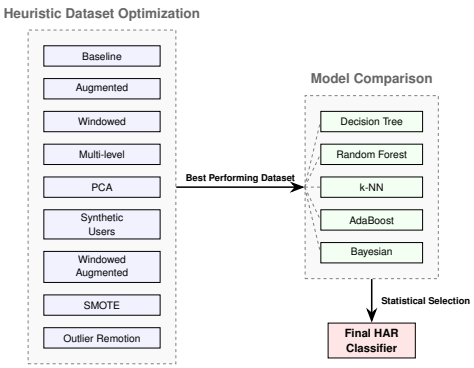


Figure 5. Overall Pipeline

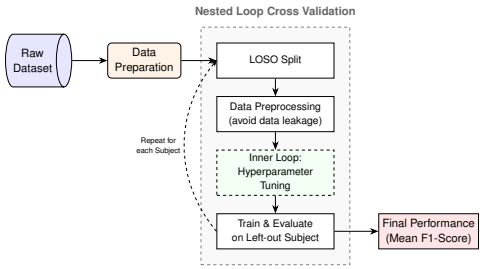


Figure 6. General ML Pipeline

3.1 Preprocessing

Before model training, the raw dataset was transformed into a representation suitable for classification. The pipeline combined categorical encoding, accelerometer-based feature

engineering, temporal windowing, and model-specific scaling. All transformations that could depend on data statistics were fitted inside the training folds to avoid leakage. Data Preparation and Preprocessing refer to the Figure 6.

Data Preparation

- **Categorical encoding**: categorical variables were transformed with one-hot encoding.
- **Signal features**: magnitude, roll, pitch, and yaw were generated from the accelerometer axes to describe movement intensity and orientation.
- **Temporal windowing**: samples were grouped into 8-sample windows with step 7; each window was described by mean, variance, minimum, and maximum values and labelled with the majority class when at least 60% of samples agreed.
- **Synthetic users**: additional users were generated as mixtures of real subjects. Note: this is a biased experiment.

Data Preprocessing
The following procedures prevent data leakage.

- **k-NN scaling**: numerical variables were standardized with Z-score normalization because k-NN depends on distances.
- **Discretization**: average magnitude values were binned using thresholds learned only on the training data.
- **Feature reduction**: Principal Component Analysis (PCA) was tested to reduce dimensionality, but it significantly decreased performance.
- **Oversampling**: minority classes were oversampled with Synthetic Minority Over-sampling Technique (SMOTE) only in the training folds to avoid test contamination.
- **Outlier removal**: outlier filtering was tested as an additional dataset variant.

3.2 Feature Selection and Engineering

Several dataset variants were compared with a Decision Tree proxy model to select the representation used in the final model comparison. Table 4 summarizes the tested variants described in Section 3.1.

Name	Description	Average F1-scores
Baseline	Raw dataset has been used without any modification	0.3783
Augmented	Dataset + Mag, Roll, Yaw and Pitch $\forall$ accelerometer	0.4290
Windowed	Windowed data samples	0.6739
Multi-level	Subsequent activities distinctions	0.4595
PCA	Dataset reduced through PCA	0.4153
Synthetic Users	New data were generated to improve generalization	0.5993
Win-Augm	Combination of augmented and windowed data	0.6005
SMOTE	Oversampling of minority classes	0.6182
Outlier Remotion	Dataset standardiziation through outlier remotion	0.5640

Table 4. Dataset experiments on tree-based models

The windowed dataset achieved the best average F1-score and was therefore selected for the

final comparison. SHAP importance analysis on this representation (Figure 7) was then used for feature selection; age, weight, and height were removed because they ranked among the least informative variables and to increase generalization capabilities on a 4 users dataset.

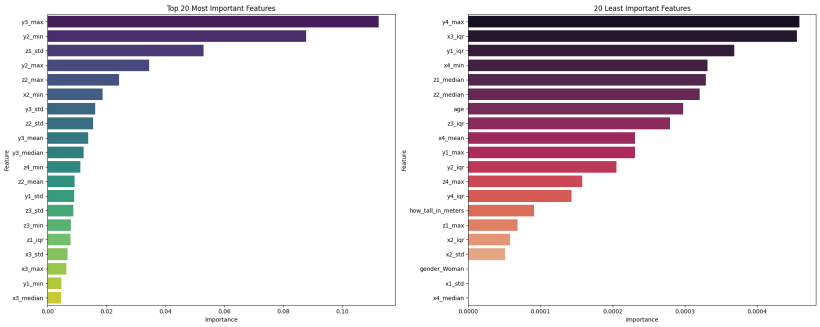


Figure 7. Feature importance scores.

3.3 Model Selection and Cross-Validation

Five classifiers were compared on the selected Windowed dataset: Decision Tree, Random Forest, AdaBoost, Gaussian Naive Bayes, and k-Nearest Neighbors. This set covers interpretable tree baselines, ensemble models, a probabilistic benchmark, and a distance-based non-parametric method. Table 5 summarizes the tuning strategy, validation scheme, and metrics used for each model.

Evaluation used nested cross-validation. The outer loop followed a Leave-One-Subject-Out (LOSO) scheme to estimate generalization to unseen users, while the inner loop used stratified folds for hyperparameter tuning. All preprocessing steps were fitted only on the training portion of each fold, as described in Section 3.1, to prevent data leakage. Weighted F1-score was used for both validation and testing because the transition classes are less frequent than static and walking classes. Figure 8 reports the score boxplots across candidate models.

Model	HP tuning	CV	Val.	Test
DT	Random	Nested	F1-w	F1-w
RF	Random	Nested	F1-w	F1-w
AB	Random	Nested	F1-w	F1-w
kNN	Grid	Nested	F1-w	F1-w
NB	Random	Nested	F1-w	F1-w

Table 5. Comparison of candidate models.

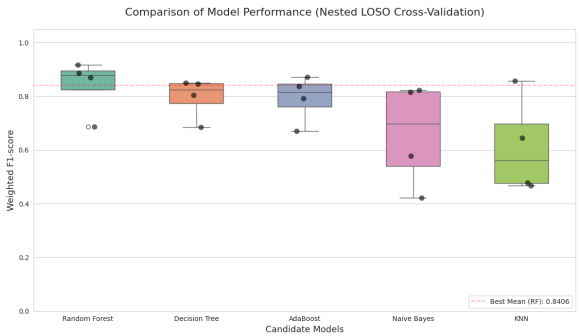


Figure 8. LOSO Cross-validated scores by model.

3.4 Hyperparameter Optimization

Hyperparameters were selected inside the inner cross-validation loop using weighted F1-score. Random Search was used for Decision Tree, Random Forest, AdaBoost, and Naive Bayes to explore broad search spaces with controlled cost. Grid Search was used for k-NN because its search space was smaller. Hyperparameter search space is described in Table 6. The best overall configuration was obtained by Random Forest; the corresponding tuning results are shown in Figure 9.

Model	Hyperparameter search space
RF	`n_estimators`: {100,200,300,500}; `max_depth`: {None,10,20,30,50}; `min_samples_split`: {2,5,10}; `min_samples_leaf`: {1,2,4}; `max_features`: {sqrt,log2}; `bootstrap`: {True,False}
DT	`criterion`: {gini,entropy}; `max_depth`: {None,5,10,20,30}; `min_samples_split`: {2,5,10,20}; `min_samples_leaf`: {1,2,4,8}; `max_features`: {None,sqrt,log2}
AB	`n_estimators`: {50,100,200,500}; `learning_rate`: {0.01,0.05,0.1,0.5,1.0}; `algorithm`: {SAMME,SAMME.R}
k-NN	`n_neighbors`: {3,5,7,9}; `weights`: {uniform,distance}; `metric`: {euclidean,manhattan}
NB	`var_smoothing`: $10^x, x \in \text{logspace}(0, -9, 100)$

Table 6. Hyperparameter search spaces.

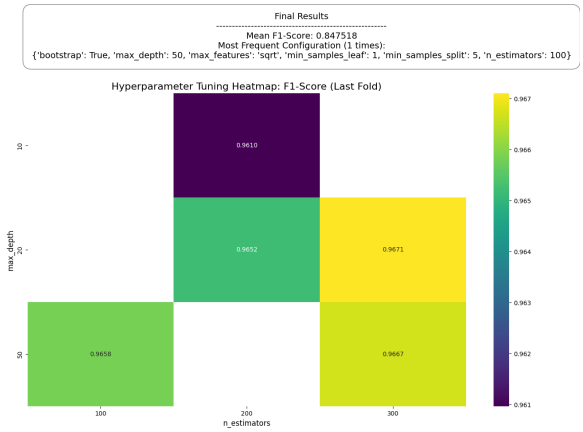


Figure 9. Hyperparameter optimization results.

3.5 Statistical Tests

To evaluate the robustness of the results and determine if the performance of the best-performing model, `Random Forest`, was significantly better than the other candidates, two statistical approaches were employed.

3.5.1 Wilcoxon Signed-Rank Test

A Wilcoxon signed-rank test was performed to compare the paired weighted F1-scores across the 4 outer LOSO folds. This non-parametric test was chosen because it does not assume a normal distribution of the scores. The `Random Forest` model was compared individually against each other model.

The results of the test are summarized in Table 7.

Table 7. Wilcoxon Signed-Rank Test results (Random Forest vs. Others)

Comparison	Statistic (W)	p-value	Significant ($p < 0.05$)
RF vs `Decision Tree`	0.0	0.125	No
RF vs `AdaBoost`	0.0	0.125	No
RF vs `Naive Bayes`	0.0	0.125	No
RF vs `KNN`	0.0	0.125	No

Interpretation: Due to the small sample size ($n = 4$ users/folds), the Wilcoxon test is mathematically unable to reach a significance level of $p < 0.05$, even when one model outperforms the other in every single fold (as indicated by the Statistic $W = 0.0$). While the results show a consistent trend favoring `Random Forest`, they do not reach formal statistical significance.

3.5.2 Probability of Superiority

Given the limitations of the p-values with a small number of subjects, a Bayesian-inspired approach using the Student's t-distribution was used to calculate the *Probability of Superiority*. This metric estimates the probability that `Random Forest` is truly better than the competing models.

- **RF vs** `Naive Bayes`**:** 96.0% probability of superiority.
- **RF vs** `KNN`**:** 94.7% probability of superiority.
- **RF vs** `Decision Tree`**:** 86.8% probability of superiority.
- **RF vs** `AdaBoost`**:** 86.6% probability of superiority.

3.6 Final Model Training

After identifying these optimal settings, the final classifier was retrained on the entire available dataset to maximize the information used for the final deployment model.

Generalization was assessed through the Nested LOSO Cross-Validation framework described in the previous sections. By using the outer LOSO loop, we simulated the deployment of the system on a new, unseen subject. The consistency of the F1-scores across different folds (users) served as the primary metric for generalization.

4. Results and Discussion

A previous study on the same PUC-Rio dataset reported higher performance using standard k-fold validation [4]. This comparison must be interpreted carefully. In k-fold validation, samples from the same subject can appear in both training and test folds, so the classifier may exploit subject-specific patterns. In contrast, the LOSO protocol used here evaluates the model on a completely unseen subject. Therefore, the lower scores obtained in this work provide a more realistic estimate of deployment performance.

4.1 Performance Evaluation

Figures 10 and 11 show that static and cyclic activities are recognized more reliably than transition classes. Sitting, standing, and walking last longer and produce more stable acceleration patterns, while *sittingdown* and *standingup* are short movements with similar window statistics when direction is not explicitly encoded. This confirms the limitation already observed in the Multi-Level experiment discussed in Section 3.2.

The most relevant errors concern transition classes. In healthcare or assisted-living scenarios, missing a transition can hide a meaningful posture change, while a false transition detection can generate unnecessary alerts or overestimate user activity. The LOSO folds also reveal strong subject dependence: Katia is the most difficult held-out user. She was at start considered as the most standard user, and her remotion from training set decreased generalization capabilities. However the experiment excluding her slightly improved performance. This suggests that she is the most distinctive subject in the dataset, consistent with Table 2, and confirms that cross-subject generalization is limited by the small number of users.

4.2 Model Explainability

During exploratory analysis, sensors 2 and 3 (left thigh and right ankle) emerged as the most informative devices, capable of capturing differences between the classes. The global SHAP values in Figure 12 shows that this summarization is too simplistic: waist sensor (1) turned out to be effective, whereas the one on the upper arm confirms expectations. The low importance of some demographic variables and engineered orientation features is described in Section 3.2: not all additional variables improve generalization under LOSO validation.

The local explanation in Figure 13 confirms gloabal features analysis and the imbalanced limitation: when transition windows have feature values close to static postures, the model tends to favor the more frequent classes. In fact the starting probability of minority class is significantly lower than others.

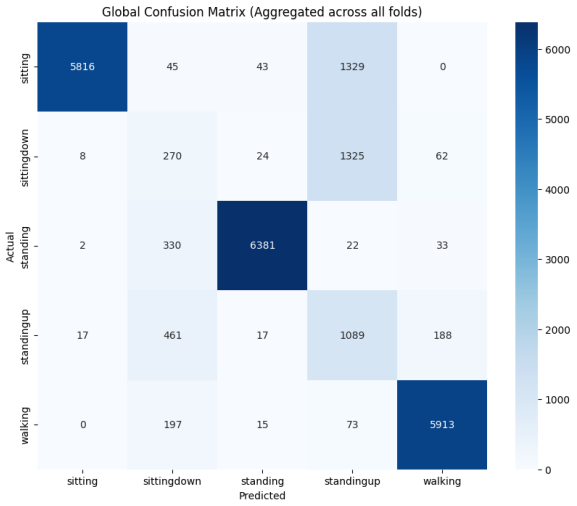


Figure 10. Primary performance result on the test set.

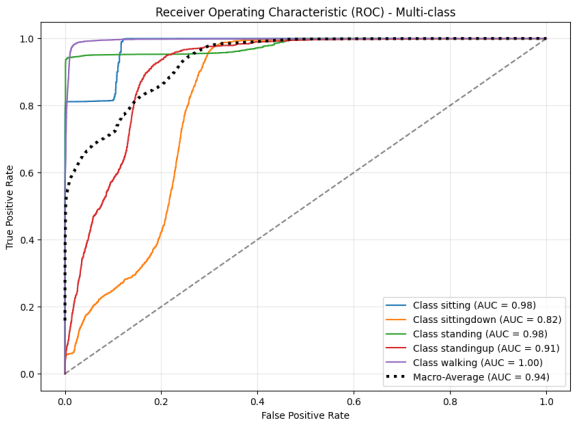


Figure 11. Secondary performance metric on the test set.

4.3 Error Analysis

Most errors are concentrated between *sittingdown*, *standingup*. This pattern is understandable because mean, variance, minimum, and maximum values capture movement intensity, but they do not fully encode temporal direction. As a result, a window corresponding to a subject moving from standing to sitting can resemble a window corresponding to the opposite transition. The feature profile of misclassified transition samples is therefore ambiguous: acceleration magnitude and variability indicate that motion is occurring, but they do not always identify whether the user is starting from sitting or standing. Class imbalance further affects predictions. Since static classes are more frequent and often share similar values with transition windows, the classifier has a tendency to assign uncertain samples to the dominant classes.

A practical way to reduce these errors would be to include temporal context beyond isolated windows. In real deployments, the previous activity is usually known: if the user was standing immediately before the transition, *sittingdown* becomes more likely; if the user was sitting, *standingup* becomes more likely. Sequential models could therefore improve recognition of the two most difficult classes.

5. Conclusions

This project addressed the classification of human postures and movements from wearable accelerometer data. The main objective was met: a complete HAR pipeline was developed, including preprocessing, nested LOSO validation, model comparison and SHAP-based explainability. Among the tested classifiers, Random Forest achieved the best overall performance, confirming the usefulness of ensemble tree models for this task.

The experiments also showed that windowed representation improved performance compared with the raw dataset, while additional orientation-based features and dimensionality reduction did not provide clear benefits. The LOSO protocol highlighted the main limitation of the system: generalization to unseen users remains difficult with only four subjects and an $8\,\mathrm{Hz}$ sampling frequency. The most problematic classes were *sittingdown* and *standingup*, because window statistics capture movement intensity but not always the direction of the transition.

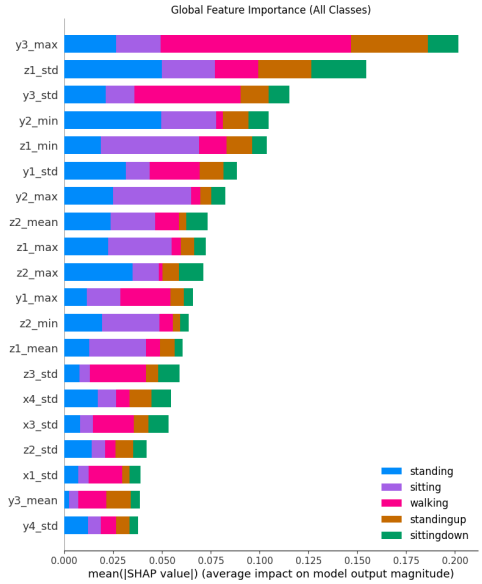


Figure 12. Global explainability: mean $|\mathrm{SHAP}|$ values.

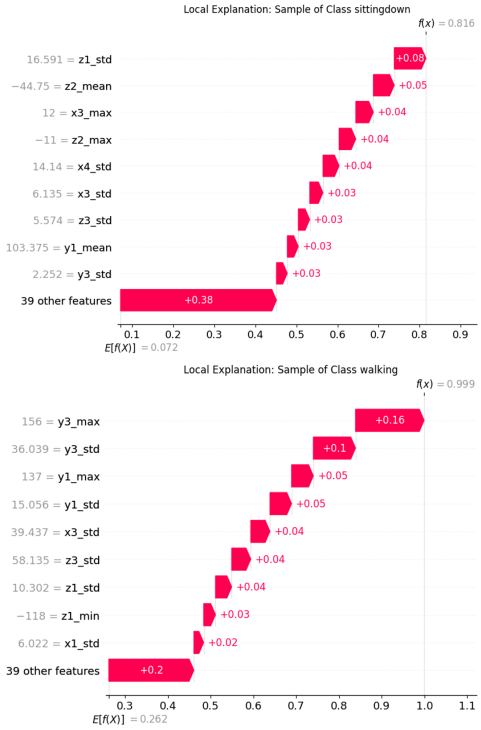


Figure 13. Local explainability: SHAP plot for a representative sample.

Future works should increase the number and diversity of subjects, collect more realistic activity sequences, and include temporal context from previous activities. Sequential models or transition-aware post-processing could help distinguish opposite movements such as *sitting-down* and *standingup*. Federated learning is also a promising direction to build more general models while avoiding the direct sharing of raw personal sensor data.

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